

# WILL LIM WEN LONG

Senior Data Scientist | IMDA SG Digital Scholar

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## PROFESSIONAL SUMMARY

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Senior Data Scientist with 3+ years building production ML systems at PwC, specialising in LLMs, RAG, real-time data pipelines, and recommendation systems. Proven track record delivering US\$1.2M in client revenue through Agentic Graph RAG solutions. Built Willi — a live RLHF fine-tuning system using Kafka streaming, DPO optimisation, and LoRA adapters — demonstrating end-to-end ML engineering from training to production deployment. AWS ML Specialty certified. Currently pursuing NUS Master of Computing (AI), cGPA 4.75/5.0.

## WORK EXPERIENCE

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### PricewaterhouseCoopers

Jan 2025 – Present

Senior Data Scientist | Singapore

- Led architecture and delivery of Agentic Graph RAG solutions (LangChain, LlamaIndex, PyTesseract), directly generating US\$1.2M in new client revenue.
- Engineered hyperparameter-tuned chunking and embedding strategies for vector DB retrieval, improving splitting accuracy by 20% over baseline and directly improving LLM answer quality in production.
- Custom-trained lightweight U-Net models for document image segmentation, improving extraction accuracy by 10% and reducing downstream OCR errors in client-facing pipelines.
- Drove end-to-end ML lifecycle across financial services and public sector clients: scoping, training, evaluation, deployment, and C-suite communication.

### PricewaterhouseCoopers

Jun 2023 – Jan 2025

Data Scientist | Singapore

- Developed multi-variate time-series forecasting models (ARIMAX, XGBoost, LightGBM) for inventory demand planning, reducing MAPE by 20% across 15+ SKU categories.
- Re-architected RAG inference pipelines with asyncio and multiprocessing, achieving 80% runtime reduction — enabling real-time response latency for client-facing applications.
- Built automated web scraping and data mining pipelines (Selenium, SQL, factor analysis) for market intelligence and competitive whitespace identification across 3 countries.
- Collaborated with product managers, engineers, and business stakeholders to translate analytical findings into product decisions.

### Diners Club International

Jan 2023 – Jun 2023

Data Analyst | Singapore

- Deployed automated data crawling, ingestion, and reporting pipelines, saving 30+ minutes of manual work daily and improving data freshness for leadership dashboards.
- Built and maintained complex PostgreSQL/MySQL queries and Cron Job workflows supporting real-time reporting for business stakeholders.

## FEATURED PROJECT

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### Willi — Live RLHF-Powered AI Clone

2026

Personal Project | askwilli.dev | github.com/wenlong96/streaming-rl-llm |

- Built a production RLHF pipeline from scratch: fine-tuned Qwen 2.5 3B Instruct with LoRA adapters and DPO (Direct Preference Optimisation) — the same alignment technique used at OpenAI and Anthropic.
- Engineered real-time feedback streaming architecture with Redpanda (Kafka-compatible): admin corrections are published as DPO training pairs, consumed live by a training loop, and model weights update without server restart.
- RAG retrieval layer using ChromaDB, LangChain, and sentence-transformers; served via FastAPI with token-streaming responses and Streamlit frontend deployed at askwilli.dev via Cloudflare Tunnel.
- Demonstrates full-stack ML engineering: data ingestion, streaming infrastructure, online fine-tuning, LoRA adapter management, and deployment — fully built without managed ML services.

## SKILLS & TECHNOLOGIES

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**Languages:** Python, SQL, R

**ML / Deep Learning:** Transformers, LoRA/PEFT, DPO/RLHF, RAG, GNN, CNN, RNN/LSTM, Time Series, XGBoost, LightGBM, Bayesian Inference

**LLM / AI Frameworks:** PyTorch, TRL, HuggingFace Transformers, LangChain, LlamaIndex, ChromaDB, sentence-transformers

**Data & Streaming:** Kafka / Redpanda, asyncio, Selenium, PostgreSQL, MySQL, Pandas, NumPy

**Deployment & Infra:** FastAPI, Streamlit, Docker, AWS EC2, Cloudflare Tunnel, Git, Azure DevOps

**Visualisation:** PowerBI, Tableau, Plotly

**EDUCATION**

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| <b>National University of Singapore</b><br><i>Master of Computing — Artificial Intelligence Specialisation   cGPA: 4.75 / 5.0   Singapore</i>  | <i>Jan 2025 – Jun 2027</i> |
| <b>Singapore Management University</b><br><i>B.Sc. Economics, Data Science &amp; Analytics   Magna Cum Laude   GPA: 3.71 / 4.0   Singapore</i> | <i>Aug 2019 – May 2023</i> |
| <b>Singapore Polytechnic</b><br><i>Diploma in Banking and Finance   GPA: 3.52 / 4.0   Singapore</i>                                            | <i>Apr 2013 – Apr 2016</i> |

**AWARDS & CERTIFICATIONS**

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|-------------------------------------------------------------------------------|-------------|
| • IMDA Singapore Digital Scholarship (Postgraduate) — Artificial Intelligence | <i>2025</i> |
| • AWS Certified Machine Learning — Specialty                                  | <i>2024</i> |
| • PwC Spot Award — Top Performer in Division                                  | <i>2024</i> |
| • SMU School of Economics Dean's List                                         | <i>2022</i> |